

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE205	DATA STRUCTURES AND ALGORITHMS	3	0	2	4	
Course Weightage	ATT: 5 CA: 50 ETP: 45					
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: understand the time and space complexity of programs and data-structures.

CO2 :: illustrate the importance of Linked List in context of real world problems

CO3 :: differentiate the Stack and Queue data structures for problem solving

CO4 :: use of recursion in iteration process and tree data structure

CO5 :: apply heap operations and hashing techniques for efficient data storage, retrieval, and collision resolution.

CO6 :: identify and integrating emerging data structures and algorithmic ensuring their ability to stay with technological advancements and maintain professional growth in a dynamic field

TextBooks (T)			
Sr No	Title	Author	Publisher Name
T-1	DATA STRUCTURES	SEYMOUR LIPSCHUTZ	MCGRAW HILL EDUCATION
Reference Books (R)			
Sr No	Title	Author	Publisher Name
R-1	DATA STRUCTURES AND ALGORITHMS	ALFRED V. AHO, JEFFREY D. ULLMAN AND JOHN E. HOPCROFT	PEARSON

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://www.geeksforgeeks.org/data-structures/	Every data structure explained with the help of demo program
RW-2	http://www.cs.usfca.edu/~galles/visualization/Algorithms.html	Data structures visualization
RW-3	https://www.tutorialspoint.com/data_structures_algorithms/index.htm	Tutorials give a clear understanding of concepts in easy and simplified manner

Software/Equipments/Databases		
Sr No	(S/E/D) (only if relevant to the course)	Salient Features
SW-1	Dev C++ /Borland C++/Turbo C++	IDE to implement data structures in convenient and faster way

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Introduction(Basic Data Structures.)	T-1 R-1	RW-1 RW-3	Lecture 0, Introduction to data structures	Remembering basic concepts and algorithmic notations used in data structures, complexity of algorithms	Lecture cum demonstration, brain storming	Keeping files in a folder, directory structure
	Lecture 2	Introduction(Basic Concepts and Notations)	T-1 R-1	RW-1 RW-3	Applications of different types of data structures and time space trade off	Understanding how to apply suitable data structure for given application	Lecture cum demonstration, brain storming	Moves on chess board
		Introduction(Complexity analysis: time space and trade off)	T-1 R-1	RW-1 RW-3	Applications of different types of data structures and time space trade off	Understanding how to apply suitable data structure for given application	Lecture cum demonstration, brain storming	Moves on chess board
		Introduction(Omega Notation)	T-1 R-1	RW-1 RW-3	Applications of different types of data structures and time space trade off	Understanding how to apply suitable data structure for given application	Lecture cum demonstration, brain storming	Moves on chess board
		Introduction(Theta Notation)	T-1 R-1	RW-1 RW-3	Applications of different types of data structures and time space trade off	Understanding how to apply suitable data structure for given application	Lecture cum demonstration, brain storming	Moves on chess board

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Week 1	Lecture 2	Introduction(Big O notation)	T-1 R-1	RW-1 RW-3	Applications of different types of data structures and time space trade off	Understanding how to apply suitable data structure for given application	Lecture cum demonstration, brain storming	Moves on chess board
	Lecture 3	Arrays(Linear arrays: memory representation)	T-1 R-1	RW-1 RW-3 SW-1	traversal, insertion and deletion operation in arrays	Understanding manipulation of elements in an array	Lecture cum demonstration, brain storming	Placing a chair in between a row of chairs
		Arrays(Array operations: traversal, insertion, deletion, sorting, searching and merging and their complexity analysis.)	T-1 R-1	RW-1 RW-3 SW-1	traversal, insertion and deletion operation in arrays	Understanding manipulation of elements in an array	Lecture cum demonstration, brain storming	Placing a chair in between a row of chairs
Week 2	Lecture 4	Arrays(Linear arrays: memory representation)	T-1 R-1	RW-1 RW-3 SW-1	traversal, insertion and deletion operation in arrays	Understanding manipulation of elements in an array	Lecture cum demonstration, brain storming	Placing a chair in between a row of chairs
		Arrays(Array operations: traversal, insertion, deletion, sorting, searching and merging and their complexity analysis.)	T-1 R-1	RW-1 RW-3 SW-1	traversal, insertion and deletion operation in arrays	Understanding manipulation of elements in an array	Lecture cum demonstration, brain storming	Placing a chair in between a row of chairs
	Lecture 5	Sorting and Searching (Searching: Linear Search and Binary Search)	T-1 R-1		Introduction to Search Algorithms	Differences between linear and binary search in terms of their approach, efficiency, and use cases	Lecture cum demonstration, brain storming	Searching a record from the group of records
	Lecture 6	Sorting and Searching (Bubble sort)	T-1 R-1	RW-1 RW-2 SW-1	Sorting elements of array	Analyzing how to apply appropriate sorting algorithm at appropriate place	Lecture cum demonstration, brain storming	Making students sit according to roll-nos. in a row
		Sorting and Searching (Insertion sort)	T-1 R-1	RW-1 RW-2 SW-1	Sorting elements of array	Analyzing how to apply appropriate sorting algorithm at appropriate place	Lecture cum demonstration, brain storming	Making students sit according to roll-nos. in a row
		Sorting and Searching (Selection sort)	T-1 R-1	RW-1 RW-2 SW-1	Sorting elements of array	Analyzing how to apply appropriate sorting algorithm at appropriate place	Lecture cum demonstration, brain storming	Making students sit according to roll-nos. in a row
	Lecture 7	Linked Lists(Introduction)	T-1 R-1	RW-1 RW-3 SW-1	Introduction to linked list and its creation	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation

Week 3	Lecture 7	Linked Lists(Memory representation)	T-1 R-1	RW-1 RW-3 SW-1	Introduction to linked list and its creation	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation
		Linked Lists(Allocation)	T-1 R-1	RW-1 RW-3 SW-1	Introduction to linked list and its creation	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation
	Lecture 8	Linked Lists(Traversal)	T-1 R-1	RW-1 SW-1	Introduction to linked list and its creation	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation
		Linked Lists(Insertion)	T-1 R-1	RW-1 SW-1	Introduction to linked list and its creation	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation
	Lecture 9	Linked Lists(Traversal)	T-1 R-1	RW-1 SW-1	Introduction to linked list and its creation	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation
		Linked Lists(Insertion)	T-1 R-1	RW-1 SW-1	Introduction to linked list and its creation	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation
Week 4	Lecture 10	Linked Lists(Deletion)	T-1 R-1	RW-1 SW-1	Algorithm to traverse and delete nodes in linked list	Understanding linked representation of memory and dynamic allocation	Lecture cum demonstration, brain storming	Chain, slides management in presentation
	Lecture 11	Linked Lists(Header linked lists: Grounded and Circular)	T-1 R-1	RW-1 SW-1	Different variant of linked list	Analyzing modification in the design of linked list to increase efficiency	Lecture cum demonstration, brain storming	Arrangement of coaches of a train
	Lecture 12	Linked Lists(Header linked lists: Grounded and Circular)	T-1 R-1	RW-1 SW-1	Different variant of linked list	Analyzing modification in the design of linked list to increase efficiency	Lecture cum demonstration, brain storming	Arrangement of coaches of a train
Week 5	Lecture 13	Linked Lists(Two-way lists: operations on two way linked lists)	T-1 R-1	RW-1 SW-1	Different variant of linked list	Analyzing modification in the design of linked list to increase efficiency	Lecture cum demonstration, brain storming	Arrangement of coaches of a train
	Lecture 14				Test - Code based 1			
	Lecture 15	Stacks(Introduction: List and Array representations, Operations on stack (traversal, push and pop))	T-1 R-1	RW-1 RW-2 SW-1	Representing stack using linked list and array, push and pop operations	Understanding implementation of stack using array and linked list as per LIFO arrangement	Lecture cum demonstration, brain storming	Navigation of directory structure in windows

Week 6	Lecture 16	Stacks(Arithmetic expressions: polish notation, evaluation and transformation of expressions.)	T-1 R-1	RW-1 SW-1	Stack representation of arithmetic expressions expression using stack	Understanding prefix, infix and postfix notations preparation	Lecture cum demonstration, brain storming	Compiler design
	Lecture 17	Queue(Array and list representation)	T-1 R-1	RW-1 RW-2 SW-1	Representation of queue using arrays and linked list and operations	Understanding memory management as per FIFO arrangement	Lecture cum demonstration, brain storming	Process scheduling algorithms
	Lecture 18	Queue(operations (traversal, insertion and deletion))	T-1 R-1	RW-1 SW-1	Different versions of queue and its operations	Understanding memory management as per FIFO arrangement	Lecture cum demonstration, brain storming	Process scheduling algorithms
		Queue(Priority Queues)	T-1 R-1	RW-1 SW-1	Different versions of queue and its operations	Understanding memory management as per FIFO arrangement	Lecture cum demonstration, brain storming	Process scheduling algorithms
Week 7	Lecture 19	Queue(Deque)	T-1 R-1	RW-1 SW-1	Different versions of queue and its operations	Understanding memory management as per FIFO arrangement	Lecture cum demonstration, brain storming	Process scheduling algorithms
SPILL OVER								
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
MID-TERM								
Week 8	Lecture 22	Recursion(Introduction)	T-1 R-1	RW-1 SW-1	Introduction of recursion and problem of tower of Hanoi	Using recursive solutions for different problems	Lecture cum demonstration, brain storming	
		Recursion(Recursive implementation of Towers of Hanoi)	T-1 R-1	RW-1 SW-1	Introduction of recursion and problem of tower of Hanoi	Using recursive solutions for different problems	Lecture cum demonstration, brain storming	
	Lecture 23	Recursion(Merge sort)	T-1 R-1	RW-1 SW-1	Algorithm of merge sort using recursive method	Using recursive solutions for different problems	Lecture cum demonstration, brain storming	
	Lecture 24	Recursion(Quick sort)	T-1 R-1	RW-1 SW-1	Algorithm of Quick sort using recursive method	Using recursive solutions for different problems	Lecture cum demonstration, brain storming	
Week 9	Lecture 25	Trees(Binary trees: introduction (complete and extended binary trees), memory representation (linked, sequential))	T-1 R-1	RW-1 SW-1	Introduction to tree data structure and its memory representation	Remembering nonlinear memory arrangements and its applications	Lecture cum demonstration, brain storming	

Week 9	Lecture 26	Trees(In-order traversal)	T-1 R-1	RW-1 SW-1	Tree traversal basics: in-order and pre-order tree traversal algorithm	Understanding node to node access	Lecture cum demonstration, brain storming	
		Trees(Pre-order traversal)	T-1 R-1	RW-1 SW-1	Tree traversal basics: in-order and pre-order tree traversal algorithm	Understanding node to node access	Lecture cum demonstration, brain storming	
	Lecture 27	Trees(Post-order traversal using recursion)	T-1 R-1	RW-1 SW-1	Tree traversal basics and post-order tree traversal algorithm	Understanding node to node access	Lecture cum demonstration, brain storming	
Week 10	Lecture 28	Trees(Binary Search Tree: introduction, searching, insertion and deletion)	T-1 R-1	RW-1 SW-1	Insertion, deletion of nodes and its algorithm in binary search tree	Understanding arrangements of data after manipulations	Lecture cum demonstration, brain storming	
	Lecture 29				Test - Code based 2			
	Lecture 30	Heaps and Hashing(Heaps: Introduction, Insertion, Deletion)	T-1 R-1	RW-1 RW-2	Introduction to heap and insertion operation	Applying another requirement to make more efficient arrangement of data in nonlinear data structure	Lecture cum demonstration, brain storming	
		Heaps and Hashing(HeapSort)	T-1 R-1	RW-1 SW-1	deletion operation and heap sort	Applying another requirement to make more efficient arrangement of data in nonlinear data structure	Lecture cum demonstration, brain storming	
Week 11	Lecture 31	Heaps and Hashing(Hashing introduction: hash functions, hash table)	T-1 R-1	RW-1 RW-2	Hashing techniques and collision detection	Applying efficient searching	Lecture cum demonstration, brain storming	
	Lecture 32	Heaps and Hashing(Open hashing (separate chaining))	T-1 R-1	RW-1 RW-2 SW-1	Hashing techniques and collision detection	Applying efficient searching	Lecture cum demonstration, brain storming	
	Lecture 33	Heaps and Hashing(Closed hashing (open addressing): linear probing)	T-1 R-1	RW-1 RW-2 SW-1	Hashing techniques and collision detection	Applying efficient searching	Lecture cum demonstration, brain storming	
Week 12	Lecture 34	Heaps and Hashing(quadratic probing)	R-1	RW-1	Hashing techniques and collision detection	Applying efficient searching	Lecture cum demonstration, brain storming	

Week 12	Lecture 34	Heaps and Hashing(double hashing)	R-1	RW-1	Hashing techniques and collision detection	Applying efficient searching	Lecture cum demonstration, brain storming	
	Lecture 35	Graphs(Graph Traversal: BFS)	T-1 R-1	RW-1	Operations on graphs and shortest path algorithm	Understanding node to node movement within graphs	Lecture cum demonstration, brain storming	
	Lecture 36	Graphs(DFS)	T-1 R-1	RW-1	Operations on graphs and shortest path algorithm	Understanding node to node movement within graphs	Lecture cum demonstration, brain storming	
Week 13	Lecture 37	Graphs(Dijkstra’s Algorithm)	T-1 R-1	RW-1	Operations on graphs and shortest path algorithm	Understanding node to node movement within graphs	Lecture cum demonstration, brain storming	
	Lecture 38				Test - Code based 3			
	Lecture 39	Graphs(Bellman-Ford Algorithm)	T-1 R-1	RW-1	Operations on graphs and shortest path algorithm	Applying efficient searching	Lecture cum demonstration, brain storming	
Week 14	Lecture 40	Graphs(Bellman-Ford Algorithm)	T-1 R-1	RW-1	Operations on graphs and shortest path algorithm	Applying efficient searching	Lecture cum demonstration, brain storming	
		Graphs(Floyd Warshall Algorithm)	T-1 R-1	RW-1	Operations on graphs and shortest path algorithm	Applying efficient searching	Lecture cum demonstration, brain storming	
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:C020102 (Total 4 tasks, 2 compulsory and out of remaining 1 best out of 2 to be considered)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Test - Code based 2	NO	30	CO3, CO4
Test - Code based 3	Yes	40	CO5, CO6
Test - Code based 1	NO	30	CO1, CO2

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Programming Practice	Yes	30	CO1, CO2, CO3, CO4, CO5, CO6
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Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allotment / submission Week
Programming Practice	To evaluate progress of individual student based upon applicability of learned concepts	The students will be given approx. 132 coding problems and 90 MCQs on the selected third party platform. The questions will be framed with equal distribution from the complete syllabus. In order to qualify for programming practice marks, the student should solve at least 50% of the coding problems and 50% of MCQ questions (eligibility condition). The maximum marks out of 15 marks for which the student would be eligible for Programming Practice would be based on the Percentage of questions solved by the student Example- If a student solves 106 questions out of 132 questions (i.e. 80% of questions solved) then the student would be eligible for 80% of 15 marks which is 12 marks (round-up would be used in case of decimal values). The final marks for Programming Practice would be calculated by prorating the eligible marks for which the student is eligible (as explained in the above point) with the percentage of marks the student has scored in the proctored Coding Contests conducted as CA's along with the mandatory written test. (The final marks would be rounded up for the students). Example – If a student solves 106 questions out of 132 questions (i.e. 80% of questions solved) then the student would be eligible for 80% of 15 marks which is 12 marks (round-up would be used in case of decimal values). And if the student has scored 24 out of 30 in the other CA's i.e. 80% marks in CA, his Programming practice final marks would be 80% of 12 marks that he was eligible for which is 9.6 rounded up to 10 marks out of 15 for Programming Practice.	Individual	Online	15	1 / 14
Test - Code based 1	To evaluate progress of individual student based upon basic data structures implementation and complexity	Mix of MCQs (10 Marks) and Coding Problems (20 Marks)	Individual	Online	30	4 / 5
Test - Code based 2	To evaluate progress of individual student based upon applicability of learned concepts	Mix of MCQs (10 Marks) and Coding Problems (20 Marks)	Individual	Online	30	9 / 10

Test - Code based 3	To evaluate progress of individual student based upon applicability of learned concepts	The test will be of one hour duration and will be conducted in offline mode	Individual	Offline	30	12 / 13
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Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	Array	Program to implement insertion and deletion operations in arrays		Implementation of array
Practical 2	Searching	Program to implement different searching techniques - linear and binary search		Implementation of searching techniques
Practical 3	Sorting	Program to implement different sorting techniques – bubble, selection and insertion sort		Implementation and analyzing how to apply appropriate sorting algorithm at appropriate place
Practical 4	Linked List	Program to implement searching, insertion and deletion operations in linked list		Implementation and analysis of different operations on linked list
Practical 5	Doubly Linked List	Program to implement searching, insertion and deletion operations in doubly linked list		Implementation and analysis of different operations on doubly linked list
Practical 6	Stack	Program to implement push and pop operations in stacks using both arrays and linked list		Understanding implementation of stack using array and linked list as per LIFO arrangement
Practical 7	Queues	Program to implement enqueue and dequeue operations in queues using both arrays and linked list		Understanding implementation of queue using array and linked list as per FIFO arrangement
Practical 8	Recursions	Program to demonstrate concept of recursions with problem of tower of Hanoi		Implementation of recursions
Practical 9	Recursive Sorting	Program to implement recursive sorting techniques - merge sort, quick sort		Implementation of merge and quick sorting
Practical 10	Tree	Program to create and traverse a binary tree recursively		Implementation of binary tree
Practical 11	Binary Tree	Program to find all paths from root to leaf in a binary tree		Understanding recursive traversal for path enumeration in binary tree

Practical 12	Binary Tree	Program to find the Lowest Common Ancestor (LCA) of two nodes in a binary tree		Understanding recursive LCA determination in binary trees
Practical 13	Binary Search Tree	Program to implement insertion and deletion operations in BST		Implementation of different operations on BST
Practical 14	Binary Search Tree	Program to find the Kth smallest and Kth largest element in a Binary Search Tree		Application of in-order traversal to solve Kth order statistics in BST
Practical 15	Heaps	Program to implement insertion and deletion operations in Heaps and Heap Sort		Implementation of heap
Practical 16	Hashing	Program to implement hash table with linear and quadratic probing		Understanding collision resolution techniques in hashing
Practical 17	Graph	Program to implement graph using adjacency matrix and adjacency list		Understanding different graph representations
Practical 18	Graph Traversal	Program to implement BFS and DFS traversal on a graph		Implementation of graph traversal techniques
Practical 19	Shortest Path	Program to implement Dijkstra's algorithms		Understanding shortest path finding algorithms
Practical 20	Graph	Program to implement Floyd-Warshall algorithms		Understanding shortest path finding algorithms
Practical 21	Graph	Program to detect a cycle in a graph using DFS		Applying DFS to detect cycles in directed/undirected graphs
Practical 22	Graph	Program to detect and remove a loop in a linked list using Floyd's cycle detection algorithm		Application of Floyd's Cycle Detection in linked lists
Practical 23	SPILL OVER			
Practical 24	Spill Over			